

Fahad Bin Mohammad Hossain

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building production backend systems for multi-tenant SaaS. Designs event-driven NestJS services with TypeScript, PostgreSQL, MongoDB, Redis, and RabbitMQ, including secure JWT/OAuth authentication, RBAC, and rate limiting. Ships AI features in production: Retrieval-Augmented Generation (RAG), embeddings, vector search, and LLM tool-calling agents (OpenAI, Claude, Voyage AI) on AWS with Docker, Kubernetes, and CI/CD.

EXPERIENCE

Associate Software Engineer II, 6sense HQ Limited - Dhaka, Bangladesh Jan 2026 - Present

- Designed a plug-and-play LLM provider interface in NestJS for Charge On Site, with multi-round tool calling over live charger, session, and report APIs so admins get user-scoped fleet insights through SSE streaming chat.
- Built JWT-scoped AI conversation threads with postgres, redis caching, rolling conversation summarization, and sandboxed Python PDF generation so operators can ask about utilization, revenue, availability, and billing in natural language.
- Reduced business-logic execution errors by 13% by building scalable NestJS APIs with retry mechanisms, idempotent workflows, and async processing across interconnected modules using PostgreSQL, MongoDB, Redis, and RabbitMQ.
- Increased AI data-pipeline throughput with event-driven processing pipelines that combine Voyage AI embeddings, Claude APIs, BullMQ background workers, and RabbitMQ messaging for higher fault tolerance.
- Strengthened multi-tenant security by implementing JWT/OAuth authentication with role-based access control (RBAC) across Global / Studio / Company scopes, and distinct rate limits per tenant and per role under concurrent load.
- Cut deployment time by 20% by managing containerized CI/CD workflows with multi-stage Docker builds, AWS ECR/EKS, and GitOps rollouts that support automated rollbacks across staging and production.
- Conducted technical interviews for Intern and Junior Software Engineer roles, mentored interns, and guided juniors through the codebase, coding standards, and company rules and delivery process.

Junior Software Engineer, 6sense Technologies - Dhaka, Bangladesh Sep 2024 - Dec 2025

- Delivered 10% faster feature rollout by building modular REST APIs in NestJS, TypeScript, and PostgreSQL with Prisma, shared validation, and high unit-test coverage.
- Reduced query latency by 17% on high-traffic transactional workflows by optimizing PostgreSQL and MongoDB with targeted indexing, query refactoring, and efficient aggregation pipelines.
- Cut large file upload times by 60% by introducing S3 pre-signed URLs that let the frontend upload files in chunks outside the request-response cycle, improving reliability for large payloads.
- Improved deployment consistency by 20% by creating reusable validation, authentication, and testing components paired with modular service design and containerized CI/CD pipelines.

Backend Developer Intern, 6sense Technologies - Dhaka, Bangladesh Mar 2024 - Jun 2024

- Learned NestJS and TypeScript by implementing small REST API features under senior review, following existing module structure, DTOs, and coding standards.
- Wrote MongoDB aggregation pipelines for filtered lists and basic reporting queries, and practiced indexing and schema reading on live collections.
- Added unit tests for assigned services and learned FastAPI basics for Python REST endpoints used in internal tooling.

SKILLS

Languages	C++, JavaScript, TypeScript, Python, SQL
Backend & Frameworks	Node.js, NestJS, FastAPI, REST APIs, GraphQL, gRPC, RabbitMQ, BullMQ
Frontend	React, NextJS, TailwindCSS
Databases	PostgreSQL, MongoDB, Redis, Valkey, VectorDB (MongoDB Atlas, pgvector)
DevOps & Cloud	CI/CD, GitHub Actions, Jenkins, Docker, Kubernetes, ArgoCD, GitOps, Sentry, AWS
AI Engineering	RAG, LLM API integration, prompt engineering, OpenAI, Claude, Voyage AI
Auth & Payments	JWT, OAuth 2.0, RBAC, Stripe
Architecture	Modular monolith, microservices, event-driven design
Testing	Jest, Unit Testing, API Testing, JMeter

PROFESSIONAL PROJECTS

- Charge On Site - EV charging operations insights assistant** - NestJS, TypeScript, Anthropic Claude, SSE, DynamoDB, Redis, JWT, Python
 - Implemented a plug-and-play LLM provider interface so that the admin assistant can swap models without rewriting business logic.
 - Wired 18+ Claude tools into existing Charge On Site APIs (companies, properties, chargers, sessions, utilization, revenue, availability, billing, warranty, ratings) so operators get live, user-scoped fleet insights in chat.
 - Shipped SSE streaming chat with thread history, Redis cache, rolling context summarization, token-usage tracking, and sandboxed Python PDF/report rendering.

- **Pattern50 - AI-era startup studio platform** - NestJS, Next.js, TypeScript, MongoDB, Redis, RabbitMQ, Claude, Voyage AI, OpenAI, Stripe, Zoho Books, AWS (S3, ECR, EKS), Kubernetes, Docker, JWT/OAuth, 1Password, Google Drive, GitHub, Codacy, Jest
 - Built the multi-tenant backend from scratch (30+ NestJS modules) for studios managing companies, products, contracts, timesheets, billing, and AI knowledge assistants.
 - Implemented JWT/RBAC with Global / Studio / Company access, invite onboarding, refresh tokens, and permission-scoped APIs for portfolio management.
 - Designed a production RAG pipeline: document chunking, Voyage embeddings, MongoDB Atlas vector search, async Bull workers, and an OpenAI report generator over company knowledge.
 - Built a Claude agent with multi-round tool calling (GitHub App + Codacy) and SSE streaming chat so studios can query repos, pull requests, workflows, and code-quality signals in real time.
 - Connected Google Drive OAuth and GitHub App ingestion into AI knowledge bases, with Bull token refresh, embedding jobs, and credentials stored in 1Password.
 - Shipped Stripe billing (PaymentIntents, cards, ACH, webhooks, Socket.IO) and Zoho Books invoicing over RabbitMQ microservices; deployed with Docker to AWS ECR/EKS via GitOps CI/CD.
- **Jeter AI - AI dating, compatibility, and matchmaking platform** - NestJS, TypeScript, PostgreSQL, Prisma, pgvector, PostGIS, OpenAI, BullMQ, Redis/Valkey, Agora, Stripe, AWS (S3, SES, ECS), Docker, Jenkins, JWT/OAuth, Jest
 - The production backend was built from scratch (25+ NestJS modules) for assessments, matching, virtual speed dating, events, and payments.
 - Designed a six-dimension compatibility engine (Heart, Mind, Body, Life, Lifestyle, Deal Breakers) with weighted scoring and GPT-generated match analysis and “Compare Us” reports.
 - Implemented RAG with OpenAI embeddings, heading-aware PDF/DOCX chunking, and PostgreSQL pgvector cosine search to ground assessment summaries in a knowledge base.
 - Orchestrated virtual speed dating: pairing optimization, BullMQ-scheduled rounds, Agora live video and speech-to-text, presence heartbeats, and post-event mutual-match conversion.
 - Shipped a Stripe token economy and event booking (PaymentIntents, 3D Secure, refunds, ledger) and deployed to AWS ECS with Jenkins, health checks, and automated rollback.
- **HTV - AI deal-flow platform for Home Technology Ventures** - NestJS, TypeScript, PostgreSQL, Prisma, OpenAI, SQS, SSE, AWS (S3, SES, SQS), Attio, JWT/RBAC
 - Built deal assessments and investment memos with OpenAI file search over S3-backed vector stores, plus web search, Firecrawl, and LinkedIn MCP tools.
 - Ran long AI jobs over SQS (assessment, memo, competitor, news, feedback learning) and streamed progress to the client with SSE.
 - Synced deals from Attio CRM via webhooks and scoped access with JWT/RBAC (roles, deal, and portfolio permissions).

EDUCATION

Bangladesh University of Business and Technology

2019 – 2023

Bachelor of Computer Science and Engineering — CGPA: 3.63

Relevant Coursework: Object Oriented Programming, Operating Systems, Data Structure and Algorithm, Artificial Intelligence, Machine Learning, Computer Networks, Database, Discrete Mathematics, Computer Architecture, Microprocessor & Microcontroller etc.

PUBLICATIONS

A Hybrid Neural Network Model for Brain Tumor Detection in Brain MRI Images

Published in: 2022 IEEE 13th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)

Date of Conference: 12-15 October 2022